

(II)

Regular - Falsi method

$f(x) = 0$ given, 1. Find $f(x_0) < 0$ and $f(x_1) > 0$
i.e. $f(x_0) f(x_1) < 0$

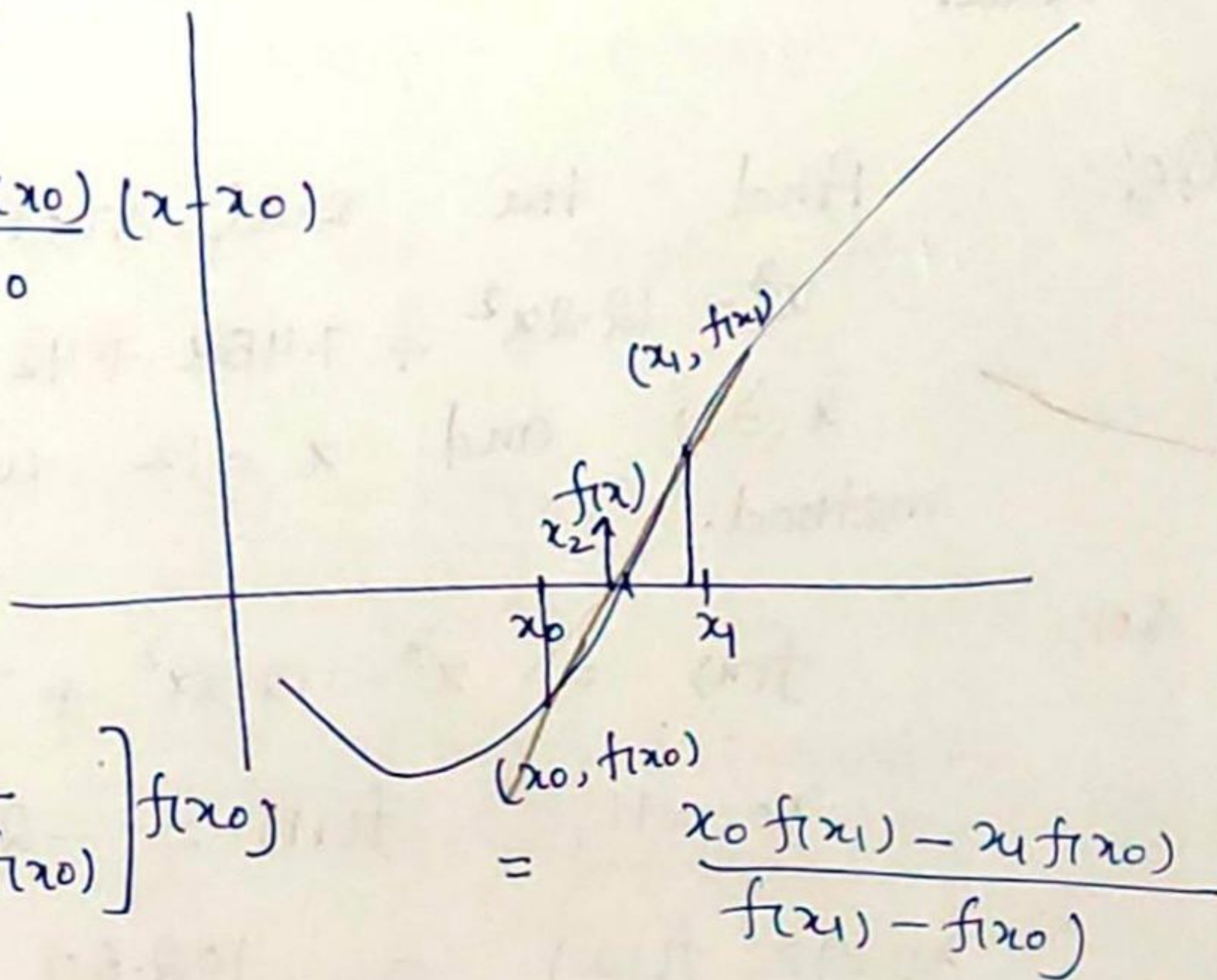
$$y - f(x_0) = \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x - x_0)$$

$$y = 0$$

$$x = x_2$$

$$x_2 = x_0 - \left[\frac{x_1 - x_0}{f(x_1) - f(x_0)} \right] f(x_0)$$

$$\text{Put } x = x_2$$



1. Given equation $f(x) = 0$

Find x_0 and x_1 s.t. $f(x_0) < 0$
and $f(x_1) > 0$ i.e. $f(x_0) f(x_1) < 0$

Root of (1) lies between (x_0, x_1)

2. Find first approximate root by
Regular falsi method

$$x_2 = x_0 - \left[\frac{x_1 - x_0}{f(x_1) - f(x_0)} \right] f(x_0)$$

3. Find $f(x_2)$ and examine its sign

3.1) If $f(x_2) < 0$ then replace $x_0 = x_2$

3.2) If $f(x_2) > 0$ " " $x_1 = x_2$

Q3) Find 2nd approximate root by

$$x_2 = x_0 - \left[\frac{x_1 - x_0}{f(x_1) - f(x_0)} \right] f(x_0)$$

Repeat these step until the approximate value.

Q4) Find the real root of the equation $x^3 - 12.2x^2 + 7.45x + 42 = 0$ lying between $x = 11$ and $x = 12$ using Regula falsi method.

Sol:

$$f(x) = x^3 - 12.2x^2 + 7.45x + 42$$

$$x_0 = 11 \quad f(11) = -21.25 < 0$$

$$x_1 = 12 \quad f(12) = 102.60 > 0$$

1st

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$= \frac{11(102.60) - 12(-21.25)}{102.60 - (-21.25)}$$

$$= \frac{1383.6}{123.85} = 11.1716$$

$$x_2 = 11.1716$$

$$f(x_2) = -3.1207 < 0$$

$f(x_1) f(x_2) < 0$ root lies between x_1 and x_2

2nd

$$x_3 = \frac{x_2 f(x_1) - x_1 f(x_2)}{f(x_1) - f(x_2)}$$

$$= \frac{11.1716 (102.60) - 12(-3.1207)}{102.60 - (-3.1207)}$$

①

$$= \frac{1183.6545}{105.7207} = 11.1960$$

$$x_3 = 11.1960 \quad f(x_3) = -0.4417$$

III rd:-

$f(x_1) f(x_3) < 0$ root lies between x_3 and x_1

$$\begin{aligned} x_4 &= \frac{x_3 f(x_1) - x_1 f(x_3)}{f(x_1) - f(x_3)} \\ &= \frac{11.1960 (102.60) - 12(-0.4417)}{102.60 - (-0.4417)} \\ &= 11.1994 \end{aligned}$$

$$x = 11.1994$$

Qo:

Using Regula-falsi method, find the root of $x \log x = 1.2$ Correct to three decimal places.

Sol:

$$f(x) = x \log x - 1.2$$

$$f(1) < 0, \quad f(2) < 0, \quad f(3) > 0$$

At least one root lies between (2, 3)

$$x_0 = 2, \quad x_1 = 3$$

$$f(x_0) = -0.59794$$

$$f(x_1) = 0.23136$$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} = \frac{2(0.23136) - 3(-0.59794)}{0.23136 + 0.59794}$$

$$x_2 = 2.721$$

$$f(x_2) = 2.721 \log 2.721 - 1.2$$

$$= -0.01709 < 0$$

the root lies between 2.721 and 3.

Now taking $x_0 = 2.721, \quad x_1 = 3$

$$f(x_0) = -0.01709$$

$$f(x_1) = 0.23136$$

$$x_3 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} = 2.74021$$

Repeat the process

$$x_4 = 2.74024$$

$$x_5 = 2.74063$$

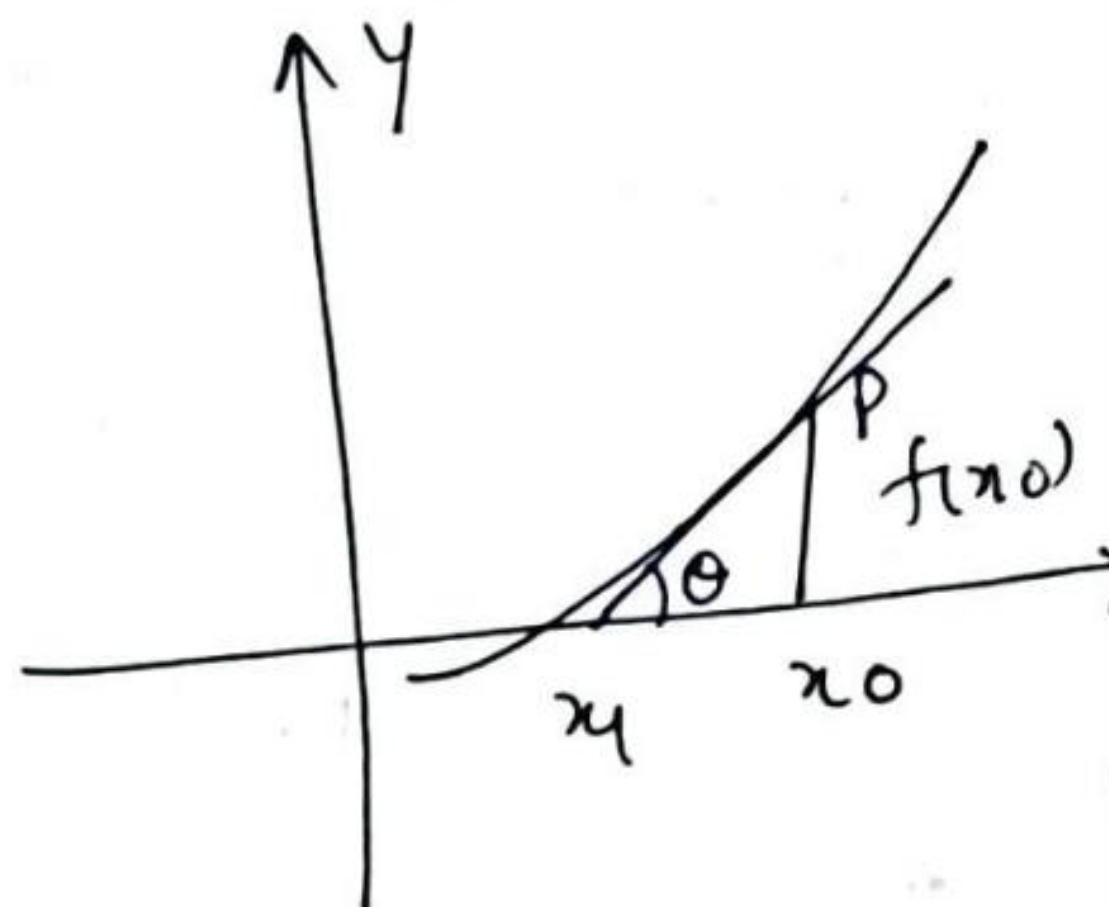
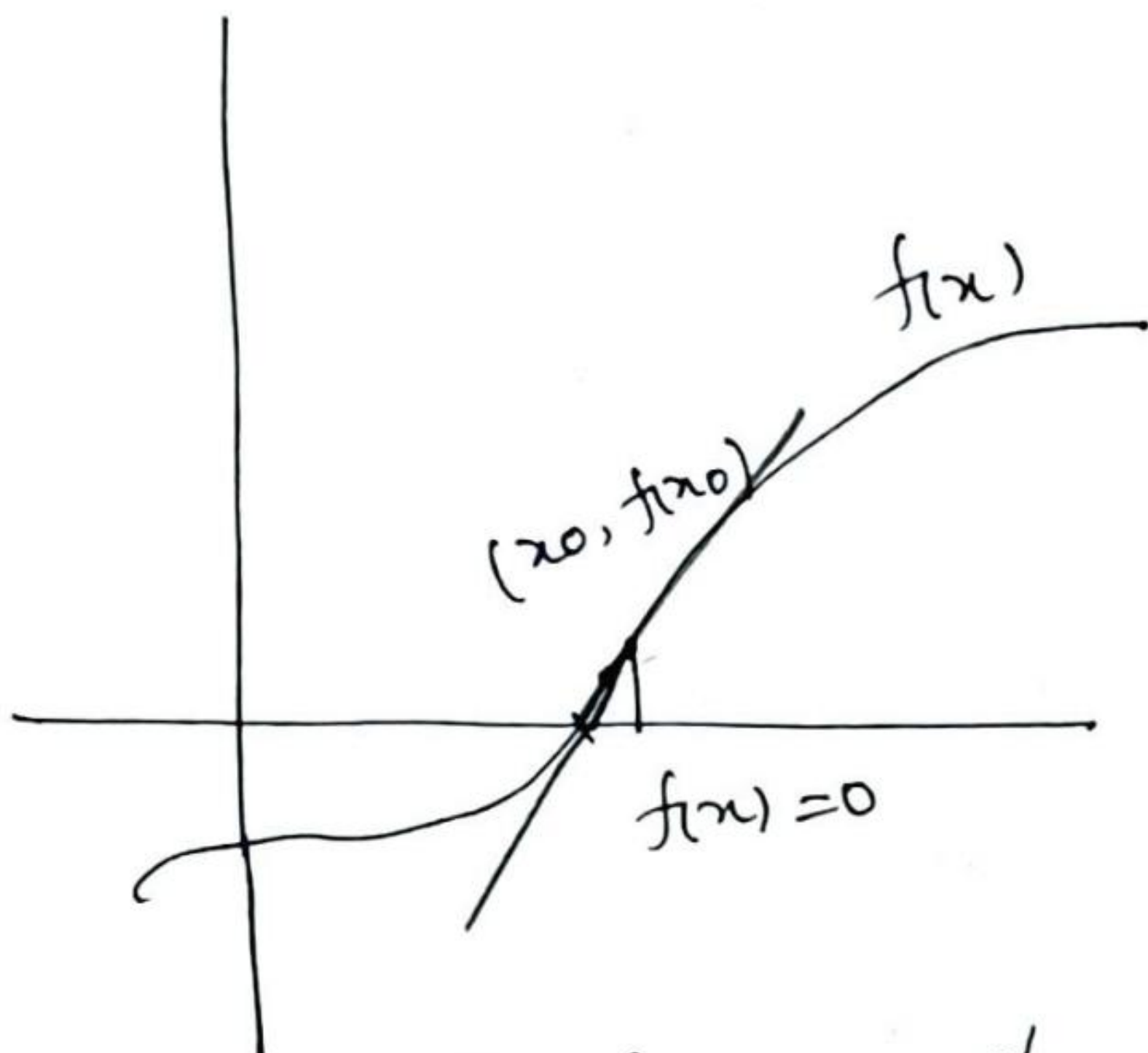
Required root is 2.7406

(IV)

Newton Raphson Method:-

(only one initial value is required) ^{or}

$$f(x) = 0$$



$$\tan \theta = \frac{f(x_0)}{x_1 - x_0}$$

$$\tan \theta = \frac{dy}{dx} = f'(x_0)$$

Put $y=0$

$$y - f(x_0) = f'(x_0)(x - x_0)$$
$$\frac{-f(x_0)}{x_1 - x_0} = f'(x_0)$$

$x = x_1$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 0, 1, 2, \dots$$

$$f(x_0 + h) = f(x_0) + hf'(x_0) + \frac{h^2}{2} f''(x_0) + \dots = 0$$

$$f(x_0) + hf'(x_0) = 0$$

$$h = -\frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 + h = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Working Rule:- 1. Given $f(x) = 0$

Find initial root x_0 s.t

$f(x_0) \sim c$ ($f(x_0)$ is near to the root

2. find $f(x_0)$ and $f'(x_0)$

3. Find approximate root by Newton

Raphson method

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Find $f(x_1)$ and $f'(x_1)$.

4. 2nd approximation

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

5.
$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

6.
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Let x_0 be an approximate root of $f(x) = 0$
and let $x_1 = x_0 + h$ be the exact
root so that $f(x_1) = 0$

$$f(x_0) + hf'(x_0) + \frac{h^2}{2} f''(x_0) + \dots$$

$$f(x_0) + hf'(x_0) = 0 \Rightarrow h = -\frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Working Rule:-

1) Given $f(x) = 0$ — (1)

Find initial root x_0 such that $f(x_0) = 0$ (i.e. x_0 is near to root of $f(x)$)

2) Find $f(x_0)$ and $f'(x_0)$

3) First approximate root by Newton-Raphson method

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Find $f(x_1)$ and $f'(x_1)$

4) 2nd approximate root

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

5) $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$

General formula

$$x_{n+1} = \overset{\text{next}}{x_n} - \frac{f(x_n)}{f'(x_n)} \quad n = 1, 2, \dots$$

Qo:-

Q.1) Determine the root of the equation

$$\cos x - xe^x = 0$$

i) Newton Raphson method

$$x_0 = 1$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$0.5403$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_{n+1} = x_n - \frac{(\cos x_n - x_n e^{x_n})}{(-\sin x_n + (1+x_n)e^{x_n})}$$

$$x_{n+1} = x_n + \frac{\cos x_n - x_n e^{x_n}}{\sin x_n + (1+x_n)e^{x_n}}$$

Put $n=0$

$$x_1 = x_0 + \frac{\cos x_0 - x_0 e^{x_0}}{\sin x_0 + (1+x_0)e^{x_0}}$$

$$x_1 = 1 + \frac{\cos 1 - e}{\sin 1 + (1+1)e} = 0.65308$$

$$f(x_1) = -0.4606$$

$n=1$

$$x_2 = x_1 - \frac{\cos x_1 - x_1 e^{x_1}}{\sin x_1 + (1+x_1)e^{x_1}}$$

$$= 0.531343$$

$$x_3 = 0.51791$$

$$x_4 = 0.51776$$

Qo! Using Newton - Raphson method, find a real root of the equation $3x = \cos x + 1$ correct to four decimal places

Sol!

$$f(x) = 3x - \cos x - 1$$

$$f(0) = -2 - \text{ve} \quad f(1) = 1.4597 \text{ +ve}$$

$f(x) = 0$ lies between 0 and 1.

$$x_0 = 0.6$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \frac{3x_n - \cos x_n - 1}{3 + \sin x_n}$$

$$x_{n+1} = \frac{x_n \sin x_n + \cos x_n + 1}{3 + \sin x_n}$$

Put $n=0$

$$x_1 = \frac{x_0 \sin x_0 + \cos x_0 + 1}{3 + \sin x_0}$$

$$x_1 = \frac{0.6 \sin 0.6 + \cos 0.6 + 1}{3 + \sin 0.6} = 0.6071$$

$n=1$

$$x_2 = \frac{x_1 \sin x_1 + \cos x_1 + 1}{3 + \sin x_1}$$

$$x_2 = \frac{0.6071 \sin(0.6071) + \cos(0.6071) + 1}{3 + \sin(0.6071)}$$

$$x_2 = 0.6071$$

$$x_1 = x_2 = 0.6071 \text{ root}$$

of the given equation

Newton's iterative formula for obtaining, inverse, square root, cube root etc :-

1. inverse:- The quantity a^{-1} can be considered as a root of the equation

$$\frac{1}{x} - a = 0$$

$$f(x) = x^{-1} - a$$

$$f'(x) = -\frac{1}{x^2}$$

$$x_{n+1} = x_n + \frac{(\frac{1}{x_n} - a)}{-\frac{1}{x_n^2}}$$

$$x_{n+1} = x_n(2 - ax_n)$$

2. Square root:- The quantity $\sqrt{a}$ can be considered as a root of the equation $x^2 - a = 0$.

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right)$$

3. Inverse square root:- The inverse square root of a is the root of the equation $\frac{1}{x^2} - a = 0$.

$$x_{n+1} = \frac{1}{2} x_n (3 - ax_n^2)$$

4. General formula for p^{th} root:-

For computing pth root of a we can solve the equation

$$x^p - a = 0$$

$$f(x) = x^p - a$$

$$f'(x) = px^{p-1}$$

$$x_{n+1} = x_n - \frac{(x_n^p - a)}{px_n^{p-1}}$$

$$x_{n+1} = \frac{(p-1)x_n^p + a}{px_n^{p-1}}$$

The reciprocal of pth root of a can be obtained by solving the equation $\frac{1}{x^p} - a = 0$ by Newton's method.

$$x_{n+1} = x_n \frac{(p+1 - ax_n^p)}{p}$$

Qo's Using Newton-Raphson method to compute $\sqrt[3]{24}$

$$x = (24)^{1/3}$$

$$x^3 - 24 = 0$$

$$f(x) = 0$$

$$f(x) = x^3 - 24$$

$$f'(x) = 3x^2$$

$$x_{n+1} = \frac{2x_n^3 - 24}{3x_n^2}$$

$$x_{n+1} = \frac{2}{3}x_n + \frac{8}{x_n^2}$$

$$x_0 = 3$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{x_0^3 - 24}{3x_0^2} = \frac{2x_0}{3} + \frac{8}{x_0^2}$$

Q06 Obtain the double root of the equation $x^3 - x^2 - x + 1 = 0$

$$f(x) = x^3 - x^2 - x + 1$$

$$f'(x) = 3x^2 - 3x - 1 \quad f''(x) = 6x - 2$$

$$x_0 = 0.8$$

$$x_0 - \frac{2f(x_0)}{f'(x_0)} = 0.8 - 2 \cdot \frac{(0.072)}{-0.68} = 1.012$$

$$x_0 - 1 \cdot \frac{f'(x_0)}{f''(x_0)} = 0.8 - 1 \cdot \frac{(-0.68)}{2.8} = 1.043$$

for the next approximation we

choose $x_1 = 1.01$

$$x_1 - 2 \frac{f(x_1)}{f'(x_1)} = 1.01 - 0.0099 = 1.0001$$

$$x_1 - \frac{f'(x_1)}{f''(x_1)} = 1.01 - 0.0099 = 1.0001$$

we can therefore conclude that there is a double root at $x = 1.0001$ which is sufficiently close to actual root $x = 1$

Since $x_0 = 3$

$$x_1 = \frac{2}{3}(3) + \frac{8}{9} = 2.889$$

$$x_2 = \frac{2}{3}x_1 + \frac{8}{x_1^2} = \frac{2}{3}(2.889) + \frac{8}{(2.889)^2}$$

$$= 1.926 + \frac{8}{8.346} = 2.8845$$

$$x_3 = \frac{2}{3}x_2 + \frac{8}{x_2^2}$$

$$x_3 = \frac{2}{3}(2.8845) + \frac{8}{(2.8845)^2}$$

$$x_3 = 1.923 + \frac{8}{8.32} = 2.8845$$

Method of multiple roots:-

$$x_{k+1} = x_k - m \frac{f_k}{f'_k}$$

$$x_{n+1} = x_n - \frac{P f(x_n)}{f'(x_n)}$$

Put $P = 1$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_0 - P \frac{f(x_0)}{f'(x_0)}, \quad x_0 - (P-1) \frac{f'(x_0)}{f''(x_0)}, \quad x_0 - (P-2) \frac{f''(x_0)}{f'''(x_0)}$$

Find the real root of equation
 $x^3 - 3x + 1 = 0$ by Newton Raphson method
correct up to 4-decimal places.

$$\therefore f(x) = x^3 - 3x + 1$$

nd x_0 :- $f(0) = 1$ +ve

$$f(1) = 1 - 3 + 1 = -1 \text{ -ve}$$

$$f(0.5) = -0.375 < 0$$

$$f(0.4) = -0.136 < 0$$

$$f(0.3) = 0.127 > 0$$

root lies between 0.3 and 0.4

$$x_0 = 0.3 \quad f(x_0) = 0.127$$

first approximate:- $f'(x) = 3x^2 - 3 \Rightarrow f'(0.3) = -2.73$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 0.3 - \frac{(0.127)}{-(-2.73)} = 0.346520$$

$$f(x_1) = 0.002048, \quad f'(x_1) = -2.639771$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.3465 - \frac{(0.002048)}{(-2.639771)} = 0.347295$$

$$f(x_2) = 0.0000167 \Rightarrow 3.5 \times 10^{-4}$$

$$f'(x_2) = -2.638356$$

$$\begin{aligned} x_3 &= 0.34729 - \frac{(0.0000167)}{(-2.638356)} \\ &= 0.34729 \end{aligned}$$